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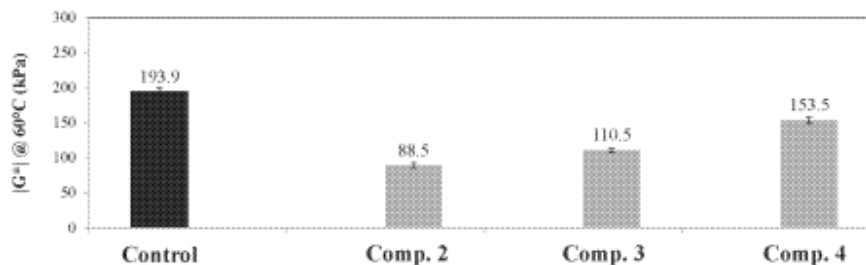
\MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <TOPICAL ASPHALT REJUVENATOR AND SEALER COMPOSITION #1> Archie Shen

Links result in unauthorized message, search US patent '20250215253' on <https://ppubs.uspto.gov/pubwebapp/static/pages/ppubsbasic.html>



|G*| change in comparison to control section after ~3-month of treatment application

- Composition 2 → Decrease of 54.3%
- Composition 3 → Decrease of 43.0%
- Composition 4 → Decrease of 20.8%

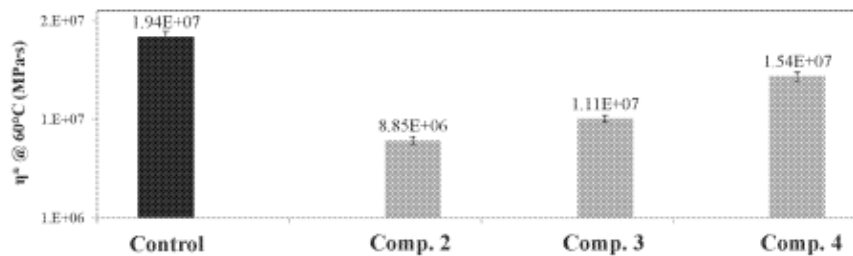
FIGURE 1

Michael Ray Leaman, Derek Crawford "Figure 1", 2025

Name: _____

Date: _____

Class: _____



η^* change in comparison to control section after ~3-month of treatment application

- > Composition 2 → Decrease of 54.3%
- > Composition 3 → Decrease of 42.9%
- > Composition 4 → Decrease of 20.7%

FIGURE 2

Michael Ray Leaman, Derek Crawford "Figure 2", 2025

Asphalt is primarily made up of three materials: sand, stone, and asphalt cement, a binder that holds them together. The binder is made up of different types of petroleum that are prone to oxidation, which causes it to harden and lose its ability to aggregate the other materials. As asphalt ages, it becomes more brittle and loses flexibility. According to Leaman and Crawford (2025), the patent's solution to this is through topical application of rejuvenating agents comprising of bitumen and other plant based oils like "soybean oil, alkyl esters from soybean oil, and terpenes used for sealing or rejuvenating" (p. 0012). The rejuvenating agents restore the original properties of the asphalt by restoring the original ratio of maltenes to asphaltenes. As asphalt ages, these two components of asphalt fall out of balance, reducing the viscoelastic properties of the binder.

This prior attempt was partly unsuccessful because while the rejuvenating agents did restore the elasticity of the binder and strengthen the existing asphalt, it did not repair the

Name: _____

Date: _____

Class: _____

damage that already existed, such as cracks and holes on the surface. In addition, there was a need for an agent that “would derive from renewable resources, would be non-toxic, quick drying, bio-based, avoid the presence of volatile organic compounds (VOCs), be capable of direct application to pavement such as by spraying, and would be capable of restoring pavement rheological properties close to those present in the original pavement” (p. 0016). While they successfully made just an agent, these many specifications would raise its price, making it less practical in comparison to other cheaper stopgap measures like simply filling in cracks and holes with sand and gravel. Gravel typically costs a dollar per square foot, while the topical rejuvenator costs 70 cents per square foot, while still not filling in the holes, the source of the problem.

The process taken to find the prior solution was by searching the US Patent Office for things relating to road repair. The US Patent Office guarantees the validity and truth of their documents.

Prior Design Attempt #2 - <Types of asphalt resilient toward environmental factors #2>

Source **Porous Pavement: Low-impact development fact sheet**

<https://extension.oregonstate.edu/catalog/pub/em-9210-porous-pavement-low-impact-development-fact-sheet>

Image

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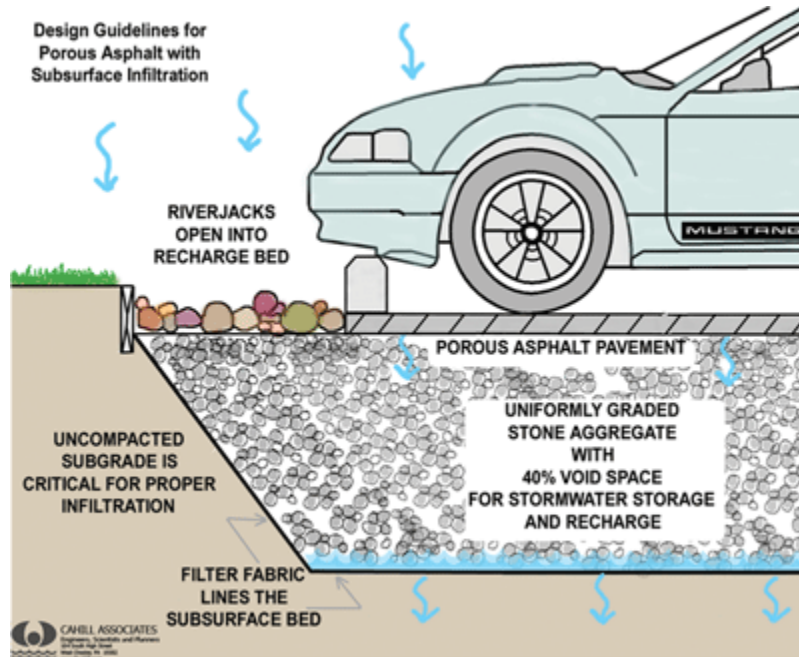


Illustration: Thomas Cahill

Porous asphalt has many names, permeable pavement, previous pavement, and it is resilient to environmental factors particularly rain as it included spaces within the asphalt to lead stormwater flow underground almost acting like a sponge. The asphalt is created in a way that no small particles remain in the substance which creates certain openings that allow stormwater to flow through.

There are three types of porous asphalt road surfaces including black pavement which has voids to let rainwater through, then there are permeable pavers which are road surfaces made of stones conjoined together through small gaps between the bricks, then there is flexible pavement which are weak flexible grids that have natural features sticking out of the holes within the grids like sand, gravel or grass.

The problem with these creations is that the voids within the pavement get clogged with overflowing water and the natural sand and rock features underneath the pavement can overflow and lead to flooding. Also if the asphalt is constantly filled with water it damages the road surface, the road also ages with UV sunlight and freezing cycles.

Name: _____

Date: _____

Class: _____

Prior Design Attempt #3 - <Asphalt Milling and Resurfacing #3>

Source

professional organization/reputable vendor and manufacturer: DUNN Co.

<https://dunnco.com/news/asphalt-milling-and-resurfacing/>



Image - Fig. II: DUNN Co, "Our Work", 2025

Milling involves using a large machine to remove the existing asphalt by using a machine with tungsten carbide teeth. The machine breaks down the pavement & collects it to be distributed in a nearby truck. After a layer of asphalt is broken down to a certain depth, the layers of asphalt below are patched if needed, cleaned, then a bonding agent & a new layer is applied, which then is rolled to become a smooth surface.

Asphalt milling & resurfacing is usually used "to address surface-level issues such as cracks, potholes, and ruts, allowing for a more efficient and cost-effective repair than full removal and replacement." (DUNN Co, 2025).

Milling & resurfacing took a lot of time, resources, & affected traffic in the area. Any features of the road such as paint & manholes would have to be adjusted & reapplied. Other solutions like binders take less time, less resources, & affect the area less. It is also more compact & easier as a solution.

Name: _____

Date: _____

Class: _____

Prior Design Attempt #4 - <Full depth patching>

Source

Asphalt Institute. CL-19: Full Depth Asphalt Patching. Asphalt Institute,
<https://www.asphaltinstitute.org/wp-content/uploads/CL-19-Full-Depth-Asphalt-Patching.pdf>



Figure IV—Spreading the mix

The Asphalt Institute, Figure IV

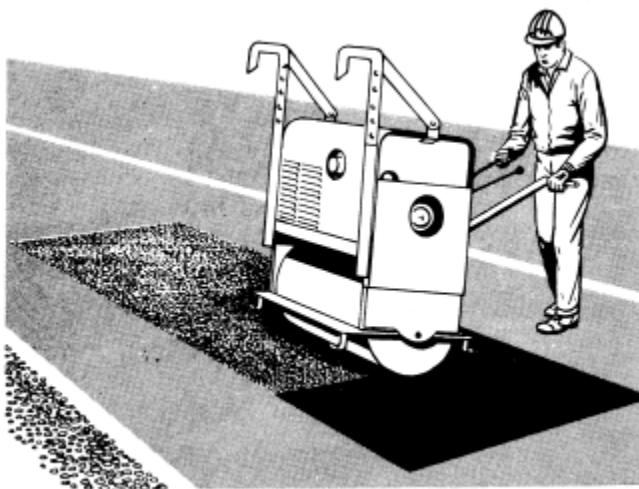


Figure V—Compacting the mix

The Asphalt Institute, Figure V

Name: _____

Date: _____

Class: _____

A full depth patching mainly consists of completely removing and redoing the damaged asphalt road. First, they remove the damaged part of the road in the direction of the traffic (so moving forward). Then, they first apply a tack coat to the sides (vertical faces). After putting the tack coat, they fill the hole up with plant mix (better quality and efficiency). Use a vibratory plate or roller to compact the patch correctly. After, check the vertical alignment of new patch, if all is well then the new patch of the road is finished.

This design works but the problem is that over time, the wear and tear on the road will just result in the same problem of cracks/ road damages, so the process needs to be repeated and there isn't a specific long lasting way to maintain the road. In addition, the patches need additional material due to the settling of the material underneath.

Prior Design Attempt #5 - <Self-Cleaning Roads Using Water Nozzles>

Source: <https://www.ijraset.com/research-paper/design-of-self-cleaning-roads-using-nozzles>

Name: _____

Date: _____

Class: _____



Image from:

<https://www.slashgear.com/1917710/how-south-korea-collects-rain-water-street-cleaning-tech/>

Detail: This design uses water nozzles directly into the road surface to create a self-cleaning system that automatically removes debris, dirt, and leaves. Sensors embedded along the roadway detect accumulation of debris and trigger the nozzles to spray water at targeted areas. This prior design attempt focused on creating self-cleaning roads by integrating water nozzles directly into the road surface. The nozzles are activated automatically by sensors that detect debris, dirt, or leaves on the road, allowing targeted cleaning without manual intervention. Water for the system can come from municipal supplies or collected rainwater, helping to reduce reliance on external resources. The design aims to improve road safety by removing slippery debris, decreasing labor costs for maintenance crews, and minimizing the use of chemical cleaning agents, making it more environmentally friendly. The nozzles are strategically spaced to maximize coverage, and the system can operate on either a scheduled cycle or a sensor-based activation, depending on debris accumulation.

Name: _____

Date: _____

Class: _____

Main details about this prior design attempt: It involved the development of self-cleaning roads using embedded water nozzles. The system relies on sensors to detect debris such as dirt, leaves, and other materials on the road surface, automatically activating the nozzles to spray water and clean affected areas. Water can be sourced from municipal supplies or harvested rainwater, aiming to reduce reliance on additional resources. The design was intended to improve road safety by keeping surfaces clear, lower maintenance labor costs, and minimize the use of chemical cleaning agents for a more environmentally friendly approach.

Analysis: While the self-cleaning road design using water nozzles presented an innovative approach to maintaining road surfaces, it had several limitations that prevented it from fully meeting design requirements. One major issue was the high water consumption, which could strain local resources, particularly in areas prone to drought. Retrofitting existing roads with the nozzle system would also require significant modifications, leading to high initial costs and potential disruptions to traffic. Additionally, the nozzles themselves are prone to clogging due to debris buildup, which can reduce the system's effectiveness and necessitate frequent maintenance.

The self-cleaning road system using embedded water nozzles was unsuccessful in fully solving the defined problem of efficient, sustainable, and low-maintenance road cleaning. Research on automated road cleaning systems reveals that high water consumption is a significant drawback, particularly in regions with limited water resources (Smith et al., 2022). Additionally, the system's dependence on proper nozzle operation created reliability issues; clogged or malfunctioning nozzles prevented consistent debris removal, leaving some areas of the road uncleaned.

